

Learning Objective

- **Gain an overview** of what AI agents are, their working, and common design patterns
- **Understand how a multi-agent system** is orchestrated to solve a business problem
- **Overview of implementing an end-to-end Multi-Agent workflow in Python** using LangGraph to coordinate agent interactions, manage shared state, enforce validation and escalation rules, and build a reliable, observable, and extensible intelligent system.

Important Points to Note

- An overview of the different functions used to set up the different components of the multi-agent system is expected in the session. A detailed explanation of the code within those functions can be skipped.
- The OpenAI API key is provided in the config file. This is only for use by the mentors and NOT to be shared with the learners.
 - Learners interested in executing the notebook on their own will be required to set up their own OpenAI API key.
- The LangSmith key needs to be generated from [this link](#) and added to the config file from the mentor's end for smooth execution during the session.

Session Structure

Time	Session Part	Activity / Step	Details
5 min	Session Overview	—	Discuss the learning objectives of the session.
20 min	Conceptual Coverage	Introduction to AI Agents	Recap of LLMs Overview of AI Agents Overview of Agent Design Patterns
85 min	Hands-on Illustration - Multi-agent System	Problem Statement & Business Context (10 min)	Set the stage with real-world relevance. Introduce the high-level flow either from the LangGraph or from the Solution Approach

Time	Session Part	Activity / Step	Details
		Import Libraries (5 min)	Import and Install Libraries
		LLM and Agent Observability Setup (10 min)	- - Load the config file with the LangSmith - Initialise the OpenAI model using the configured key
		Data Loading & Preprocessing (5 min)	- Load the dataset with pandas - Preprocess the data - Show the first few rows
		Multi-Agent Architecture Implementation (30 min)	- Describe the function and responsibilities of each agent in the system. - Illustrate how documents are retrieved through an iterative retrieval process. - Explain how the final response is grounded in the retrieved information.
		Multi-Agent System Workflow (5 min)	- Build and walk through the LangGraph workflow
		Testing the Agent Workflow using Test Case (15 Min)	- Execute the agent workflow on multiple test cases - Show the logs in the LangSmith Interface
		Business Recommendations (5 min)	Discuss business recommendations and future enhancements to the solution
10 min	Open Q&A	-	Address doubts regarding the overview of AI Agents and the use case illustrated

Power Ahead!